Item-dropping to improve in-sample Cronbach's alpha can harm true reliability: A Monte Carlo simulation

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Abstract

Cronbach's alpha is the most commonly reported metric of reliability in psychological measurement, and statistical software encourages a common post hoc revision: dropping the item whose removal most improves in-sample alpha. A Monte Carlo simulation examined what this practice does to true reliability, varying items per scale (5 to 40) and sample size (20 to 1,000), with item pools generated from a loading distribution fitted to 3,783 factor loadings from real psychological scales. Two strategies, always dropping the nominated item or dropping only when in-sample alpha improves, were evaluated against the population alpha and true reliability of the retained items. The bias in the reported alpha itself was modest, overstating true reliability only in short scales at small samples and understating it elsewhere. The decision was another matter: at the smallest scale and sample size, dropping reduced true reliability in 46.7% of the replications in which the conditional rule acted, despite requiring an apparent improvement, and almost the entire apparent gain was selection optimism. Stricter decision rules helped only in large samples: with N of 200 or more, acting only on apparent gains above .02 kept the probability of harming the scale at or below 7%, whereas at N = 20 no threshold examined made the decision safe. Because this risk depends on scale length and sample size, given loadings typical of trait scales, it is knowable before data are collected, whereas nothing inside the alpha-if-item-removed table distinguishes a genuinely weak item from a sampling artefact.

1. Introduction

Cronbach's α is the most widely reported metric of structural validity in psychology (Flake et al., 2017). It provides a lower bound of the internal consistency of a multi-item measurement scale under the assumption of tau equivalence (Cronbach, 1951) and sees wide use in many fields. However, α also has longstanding critiques of its use and interpretation, including that it is often erroneously used by researchers to test tau equivalence when it in fact assumes it (Cortina, 1993; Sijtsma, 2009). Although there have been repeated calls

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to employ other metrics such as McDonald's ω (McDonald, 1999), α is so widely used that it is useful to examine and encourage its appropriate usage.

1.1. Item dropping: measurement refinement or Questionable Measurement Practice?

Improving psychological measurement is, of course, desirable, and much work is done on scale development and refinement to reliably and validly measure constructs of interest (e.g., Clark & Watson, 1995, 2019; Jebb et al., 2021; Rodriguez et al., 2016). Selection of items from an item pool is a key step in the development of a scale (Clark & Watson, 2019), and item dropping can be an important part of ongoing scale refinement. A review of the industrial-organisational psychology literature has shown that item dropping is one of the most commonly reported forms of scale modification (Heggestad et al., 2019).

However, item dropping could also occur in ways that do not improve reliability or which undermine cumulative understanding, for example when the item dropping is not reported, or when the drop decisions are over-conditioned on the data at hand. Elson et al. (2023) makes this point directly: researchers frequently drop items they argue to be poorly performing in order to boost a scale's reliability coefficient, but without assessing whether that drop decision and associated improvement in reliability replicates in new samples. Such overfitting reduces the replicability and validity of claims. The flexible application of scale modification through, for example, item dropping, can represent a Questionable Measurement Practice or an instance of 'schmeasurement' (Flake & Fried, 2020; Hussey et al., 2025; Hussey & Hughes, 2020; Stefan & Schönbrodt, 2023).

There is currently a lack of clarity about whether, and under what conditions, item dropping from an established scale represents either a) a defendable and even desirable form of measure refinement or b) an instance of 'schmeasurement' or Questionable Measurement Practices (Flake & Fried, 2020). That is, agnostic to researchers' intentions, whether a given research practice is aligned with improving reliability and validity, and our understanding of them, or not.

This is further complicated by the mathematical basis of α itself, and the currently-unclear cumulative impact of item dropping on reliability. On the one hand, we can intuit that 'bad' items worsen α by definition, and so removing poorly performing items represents an upwards pressure on α. However, on the other hand, α is a function of the number of items in the scale, so shortening a scale exerts downward pressure on α. The cumulative impact of these upwards and downwards pressures is currently not well understood: the marginal cost of retaining a mediocre item could be smaller than the benefit of the scale being longer, or vice versa.

Relatedly, the specific method of item dropping necessarily introduces researcher degrees of freedom, and therefore Questionable Measurement Practices by definition. For example, whether items are dropped or not, how many items, how iteratively, etc. Previous simulation studies have already demonstrated that flexible application of item dropping increases the false positive rate of subsequent hypothesis tests from the nominal 5% to as high as 30% (Stefan & Schönbrodt, 2023). However, no research has examined the impact of (flexible) item dropping on the magnitude or bias of α estimates themselves. The closest analogue to this research question that we are aware of, which influenced the conceptualisation and design of the current research, is provided by Kopalle & Lehmann (1997), who simulated the impact of item dropping based on factor loadings (via exploratory

factor analysis) on the magnitude or bias of α estimates. Notably, while the authors concluded that procedures which eliminate items from scales to improve α may lead to an overestimation of a scale's α, the interpretability and credibility of this claim is undermined by a) the use of only 50 iterations per simulation, likely producing highly stable results, and b) the lack of details about what specific form of EFA was employed.

1.2. Potential influences on the research practice of item dropping

Researcher behavior is human behavior, and as such can be usefully understood as being subject to multiple sources of influence, and indeed bias. Measurement practices are, presumably, commonly influenced by researchers' scientific goal of obtaining reliable and valid measurements of their constructs of interest. The ongoing improvement of research practices is a part of science, and it is useful for the field to develop an understanding of where practices are imperfectly aligned with that scientific goal. One obvious moderator is access to sufficient training (Jans et al., 2015; Munafò et al., 2017). Researchers share the scientific goal of carefully measuring the construct of interest, but may lack sufficient training and expertise to pursue it effectively in practice.

Other forms of influence or bias are worth considering, such as the development of heuristics that become conventions in a field. For example, item dropping is plausibly influenced by our rules-of-thumb for adequate reliability, which can themselves become targets (i.e., a case of Goodhart's Law, "When a measure becomes a target, it ceases to be a good measure," Strathern, 1997). Although conventions for acceptable scale reliability are less universal for α than for other metrics (e.g., the $p < .05$ convention for statistical significance), there are commonly understood thresholds of reliability that are reflected in the literature (e.g., .70 or .80). Hussey et al. (2025) recently tested this by extracting more than 180,000 reported α values from three very large corpora to investigate the values reported, covering the general psychology literature, the industrial and organizational psychology literature, and the American Psychological Association's PsycTests database. Results demonstrated robust evidence of an excess of α values at the .70 rule-of-thumb threshold. These α values at the thresholds of conventional acceptability are best explained by publication bias and/or hacking. This may therefore represent an instance of Goodhart's Law, where the rule-of-thumb function not only as a rule for interpretation but also become a target to hit, and where reported α values become a distorted picture of actual reliability.

A second source of plausible influence on the research practice of item dropping can be found by viewing it through the lens of human-computer interaction. Many commonly used statistical analysis software packages for calculating α also provide the user with feedback on α-if-item-removed for each of the scale's items (e.g., SPSS, JASP, jamovi, and the R packages *psych* and *performance*, (Lüdecke et al., 2021; Revelle, 2024)). It is reasonable to assume that this feedback to the user influences the analytic choices made (i.e., the point of software feedback is to inform and influence subsequent user behavior), but less clear whether the improvement in in-sample α and the potential analytic flexibility this choice introduces is conducive to desirable measurement practices in a more general sense.

1.3. The current research

This article examines whether item dropping based on Cronbach's α-if-item-removed improves or worsens estimation of the true reliability of a scale. Item dropping procedures other than α-if-item-removed are of course possible, and are discussed in in the context of measure refinement Clark & Watson (2019), we focus on this specific mechanism on the basis that feedback on α-if-item-removed when calculating α is nearly ubiquitous in commonly used statistics software whereas feedback on other mechanisms is rare (e.g., SPSS, JASP, jamovi, and the R packages *psych* and *performance* (Lüdecke et al., 2021; Revelle, 2024)). Additionally, we examine the impact of a single dropped item rather than an iterative drop of multiple items on the basis that the latter requires a rapidly expanding multiverse of analytic choices whose realism is not currently known.

2. Method

We simulated the impact of item dropping using a data-generating process with parameters estimated from a large corpus of empirical self-report data from a wide range of psychological scales, so that the simulated scales resemble the reliability properties of real trait scales rather than an arbitrary parametric grid.

We adopt the ADEMP framework (Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures)for designing and reporting simulation studies for maximum interpretability and interoperability. This was originally proposed for simulation studies in biostatistics (Morris et al., 2019) and has since been adapted and promoted for psychology, including a preregistration template (Siepe et al., 2024).

2.1. Aims

The estimator under study is the sample Cronbach's α of the retained items, that is, the number the analyst actually reports, evaluated against two population targets defined for the specific items retained. The aims are:

1. Calibration and the structural gap (full pool, no dropping). To estimate the bias of sample α as an estimator of the true population α of the full item pool, which serves as a check that the simulation pipeline behaves as expected, and to quantify the baseline gap between population α and true reliability that exists before any item selection.
2. Bias in the reported reliability of the retained items. For each strategy, to estimate how much the sample α of the retained items over- or under-states (a) the true population α of those same items and (b) their true reliability.
3. What dropping actually does to the scale. To estimate the expected change from full pool to retained items on three scales (the observed sample α, the population α, and the true reliability), the optimism gap between the first two, and the proportion of replications in which dropping actually reduced true reliability.
4. How much a single study can gain or be misled. Aims 1 to 3 are long-run expectations and are silent on the tail. To address it we estimate the spread of the per-replication quantities, summarised as 95% prediction intervals, alongside every mean reported under Aims 2 and 3.

2.2. Data

For the population distribution of factor loadings, we employed a data generating process that was derived from empirical data . For this, we used *granary*, a very large aggregation of other open datasets of granular item-level psychological traits and attitudes (Hussey, 2026). Single-item measures do not yield a factor loading and were therefore excluded. For each pairing of scale and study source, participants who completed fewer than 80% of the items from that measure were excluded prior to any model fitting, using a threshold applied uniformly across datasets.

2.2.1. Fitted parametric distribution of loadings

The pooled distribution comprised 3,783 standardised single-factor loadings across 206 (scale, source) units, contributed by 196,769 distinct participants. Each loading comes from a separate single-factor confirmatory factor analysis fitted to one (scale, source) dataset rather than from a single pooled model, and because a participant may complete several scales, those 206 analyses together rest on 252,478 scale-level observations. The number of participants per unit is large on average but highly variable and strongly right-skewed: mean 1,226 (SD 2,564), median 468, interquartile range 255 to 1,027, minimum 99, maximum 19,330. Given that skew, the minimum and maximum are reported alongside the mean and standard deviation because the latter pair is on its own misleading here: the standard deviation exceeds twice the mean, 86% of units fall below the mean rather than the mean sitting near their centre, and the largest unit is nearly 200 times the smallest. The median is the more representative summary of a typical contributing scale. Scale length was similarly skewed, with a mean of 18.4 items (SD 15.0), median 13, minimum 3 and maximum 90. Loadings enter the pool unweighted, so each item counts once regardless of the sample size of the scale it came from, which is the appropriate target for a data-generating process that draws items rather than scales.

The pooled distribution was best fit, among six candidate distributions compared by AIC, BIC, and Kolmogorov-Smirnov distance, by a scaled Beta distribution obtained by rescaling the loading onto $[0, 1]$ before fitting:

$$\lambda_i \sim 2 \times \text{Beta}(11.77,\, 3.84) - 1$$

The fitted shape parameters were $\text{shape}_1 \approx 11.77$ and $\text{shape}_2 \approx 3.84$, giving a mean loading of approximately 0.51 (mean inter-item correlation ≈ 0.26) and a left-skewed distribution with an approximately 1.7% negative-loading tail. These shape parameters are estimated in a separate file (Hussey, 2026) and enter this simulation only as fixed constants. The loading distribution is not a factor of this design and is not manipulated. Two of its properties are nonetheless important for the estimands: loadings are unequal, which is what makes population α and true reliability distinct targets, and the small negative tail means some generated pools contain a genuinely α-harming item, so conditional dropping is not always chasing noise.

2.2.2. Distribution of scale lengths

The number of items per scale was recorded for each of the 206 (scale, source) units. This distribution was right-skewed, ranging from 3 to 90 items. Rather than fixing the simulated item pool size at one value, we took the 10th, 25th, 50th, 75th, and 90th

percentiles of this empirical distribution (5, 9, 13, 21, and 39 items respectively, rounded to 5, 10, 15, 20, 40 for even spacing) as the item pool sizes to simulate. The range of scale lengths studied therefore reflects how long real trait scales tend to be in published research, from short screening instruments to long, multidimensional batteries.

2.3. Data-generating process

Item-level data for one simulated scale were generated per replication from a standardised single-factor model using `lavaan::simulateData()` (Rosseel, 2012). The common factor is standard normal, item i loads λ_i on it, and its unique variance is $\psi_i = 1 - \lambda_i^2$, so that every item has unit total variance and $\mathrm{Cov}(X_i, X_j) = \lambda_i \lambda_j$ for $i \neq j$. Because the population loadings are drawn by the simulation rather than estimated from data, the true population Cronbach's α and the true reliability of the unit-weighted sum score are available in closed form for any subset of items, with no need for a separate large-N reference sample.

2.3.1. Design factors

Two factors were manipulated, crossed fully factorially:

- Item pool size (k): five levels (5, 10, 15, 20, 40 items), taken from the empirical percentiles described above.
- Sample size (N): five levels (20, 30, 50, 200, 1000), chosen to reflect a range of plausible study sizes in psychological research.

This gives $5 \times 5 = 25$ conditions. Each condition was replicated 10,000 times, for a total of 250,000 independently generated datasets. One dataset is generated per replication and both strategies are applied to it. Because both strategies remove at most one item, there is no "number of items retained" factor: how many items the conditional strategy retains is an outcome of the rule, not a manipulated input. There is no exclusion rule, since Cronbach's α is a closed-form function of the sample covariance matrix and does not require a full-rank correlation matrix, so cells with $N < k$ are retained.

2.4. Item dropping methods

Both strategies nominate the same candidate item from the sample data,

$$J^* = \arg\max_j \hat{\alpha}(-j)$$

where $\hat{\alpha}(-j)$ is the sample α of the pool with item j removed, and differ only in whether acting on that nomination is gated on an observed improvement:

- Forced drop. Remove item J^* regardless of whether $\hat{\alpha}(-J^*)$ actually exceeds the full-scale $\hat{\alpha}(F)$. The retained set is therefore: $S_{\text{forced}} = F \setminus \{J^*\}$ always.
- Conditional drop. Remove item J^* only if it improves in-sample alpha ($\hat{\alpha}(-J^*) > \hat{\alpha}(F)$), otherwise remove nothing. Therefore: $S_{\text{cond}} = F \setminus \{J^*\}$ or F.

Because the strategies share the nomination rule, comparisons between them are comparisons of the gating rule with the nomination rule held fixed. This is an intended design feature rather than a confound, and it is why the alpha-if-removed vector is computed once per replicate in the code.

2.5. Validation of the implementation

Sample α and the k leave-one-out alphas are computed directly from a single sample covariance matrix rather than by calling `psych::alpha()` once per item, which is what makes a design of this size tractable: the direct computation avoids building the large object of item statistics that `psych::alpha()` returns and that this simulation never uses, reducing the cost of a full run from hours to minutes. Substituting a hand-written implementation for the field-standard one is only defensible if the two are shown to agree, so a separate validation file, sharing the same source file of functions so that no second copy can drift out of sync, checks three distinct things before the simulation is permitted to use them.

First, agreement with `psych::alpha()`. Across 360 cases drawn from this design's own data-generating process and grid, including the 69 whose item pool contained a negatively loading item, the largest absolute discrepancy in either the full-pool α or any leave-one-out α was 1.2×10^{-15}, and the two implementations nominated the same candidate item J^* in 360 of 360 cases. Cases containing a negatively loading item are included deliberately, since neither implementation applies automatic reverse keying and such pools are the case most likely to expose a difference.

Second, the closed-form population targets. α_{pop} and ρ computed from the loadings were compared against the same quantities computed independently from the model-implied population covariance matrix $\Sigma = \lambda\lambda^\top + \text{diag}(1 - \lambda_i^2)$. The largest discrepancy was 1.3×10^{-15}. The inequality $\alpha_{\text{pop}}(S) \leq \rho(S)$ that the estimands rely on held in every case.

Third, and least obviously, that the generated data actually has the population structure the targets assume. This is not a hypothetical concern. Given `lavaan` syntax specifying loadings only, `simulateData()` leaves the residual variances free and defaults them to 1, producing items of variance $\lambda_i^2 + 1$ rather than unit variance. The population targets would then describe a different population from the one being sampled, and every bias estimand in the design would be displaced by a constant that looks exactly like finite-sample bias. The residual variances are therefore stated explicitly as $\psi_i = 1 - \lambda_i^2$, and the validation confirms this worked: generating 100,000 observations from one pool at each item pool size, the mean realised item variance departed from 1 by at most 0.0024, and the realised sample α departed from the closed-form population α by at most 0.0009, which is within Monte Carlo error at that sample size. The validation caches are keyed on a hash of the function file, so editing those functions invalidates the validation rather than allowing a stale pass to stand.

2.6. Expectations prior to running the simulation

This simulation was not preregistered. We nonetheless recorded, in the design document, what we expected each performance measure to show before the full run was executed, and we report those expectations here so that the results can be read against them rather than only in hindsight. They should be read as the authors' priors, not as a preregistered test.

We expected the full-pool calibration bias to be small and negative and to shrink toward zero as N grew, since sample α carries a known small-sample downward bias. The structural gap to be negative in every condition, since the fitted loading distribution

guarantees unequal loadings. We expected to reproduce the Kopalle & Lehmann (1997) optimism result, with the reported α biased upward relative to the population α of the retained items, and we expected that bias to be larger at larger item pool sizes, on the reasoning that more candidate items means a larger expected maximum over noise. We expected the reliability-scale bias to be negative at large N, where selection has little noise to exploit and only the structural gap remains, and possibly positive at small N, where the winner's curse might dominate – where the sign crossed, we treated as an open empirical question. We expected Pr(drop) to rise with item pool size. We made no directional prediction for the mean population-α change or the mean reliability change, because the empirical loading distribution has a negative tail, so some pools contain a genuinely harmful item whose removal helps while others pay a pure length penalty. We expected the prediction intervals to be wide relative to the means wherever selection had room to operate. Which of these held is taken up in the Discussion.

2.7. Estimands

Because the data-generating process fixes the population loadings directly, two population quantities are available in closed form for any item set S of size m, writing $S_1 = \sum_{i \in S} \lambda_i$ and $S_2 = \sum_{i \in S} \lambda_i^2$. The population Cronbach's α is

$$\alpha_{\text{pop}}(S) = \frac{m}{m-1} \cdot \frac{S_1^2 - S_2}{m + S_1^2 - S_2}$$

and the true reliability of the unit-weighted sum score, McDonald's ω, is the proportion of sum-score variance due to the common factor:

$$\rho(S) = \frac{S_1^2}{S_1^2 + \sum_{i \in S} \psi_i} = \frac{S_1^2}{S_1^2 + m - S_2}$$

By the Cauchy-Schwarz inequality, $S_1^2 \leq m S_2$, so $\alpha_{\text{pop}}(S) \leq \rho(S)$ always, with equality only under essential tau-equivalence (all loadings in S equal). Because the fitted loading distribution produces unequal loadings, this inequality is strict in essentially every generated pool: α structurally understates true reliability even at infinite sample size, before any item selection at all. This is why a congeneric data-generating process is the appropriate one for this question, since a tau-equivalent one would collapse the two targets into one by construction.

Both population targets are compared, within the same replicate, against the corresponding sample Cronbach's α, giving an exact per-replicate decomposition of the total bias against reliability:

$$\underbrace{\hat{\alpha}(S_s) - \rho(S_s)}_{\text{total bias vs. reliability}} = \underbrace{\left[\hat{\alpha}(S_s) - \alpha_{\text{pop}}(S_s)\right]}_{\text{statistical bias (sampling + selection)}} + \underbrace{\left[\alpha_{\text{pop}}(S_s) - \rho(S_s)\right]}_{\text{structural gap} \leq 0}$$

The two components can pull in opposite directions: selection pushes the reported α up relative to its own population value, while the structural gap means that population value sits below true reliability. Whether the reported α ends up over- or under-stating true

reliability, and where in the design the sign flips, is an empirical question this simulation answers.

Because J^* is a random variable across replicates, the population targets are realised (replication-specific, strategy-specific, data-dependent) estimands rather than fixed population parameters. This is deliberate: the applied question is not "is the best possible subset's reliability well estimated" but "how reliable, in truth, are the specific items this procedure just picked, and does the number I report for them tell me that?"

A second family of estimands evaluates the decision rather than the reported number:

$\widehat{\Delta}_{\text{obs}} = \hat{\alpha}(S_s) - \hat{\alpha}(F)$, the observed sample gain: the quantity the researcher sees and that, for the conditional strategy, triggers the decision. Under the forced strategy this can be negative, since the best available drop may still hurt the sample α, under the conditional strategy it is ≥ 0 by construction and exactly 0 when nothing is dropped.

$\Delta_{\text{pop}} = \alpha_{\text{pop}}(S_s) - \alpha_{\text{pop}}(F)$, the population α change for the selected scale. Removal alters the item-variance sum, the sum-score variance, and the length multiplier $\frac{m}{m-1}$ simultaneously, so the sign is a trade-off and $\Delta_{\text{pop}} \neq 0$ in general.

$\Delta_\rho = \rho(S_s) - \rho(F)$, the true reliability change, which is the quantity that actually matters for genuine measure refinement. The primary failure mode this design can detect is $\Delta_\rho < 0$ while $\widehat{\Delta}_{\text{obs}} > 0$, that is, dropping a low-consistency but valid item raises the α coefficient while degrading measurement.

The findings live in the gaps between these deltas. $\widehat{\Delta}_{\text{obs}} - \Delta_{\text{pop}}$ is the selection optimism (winner's curse), expected to be positive because J^* maximises the sample α and is therefore biased high for its own population value. It is computed pairwise within replicate, so its Monte Carlo standard error is much smaller than differencing two separate summaries would give. Because a mean Δ_ρ near zero can conceal frequent, offsetting harm, we additionally report the harm rate, $\Pr(\Delta_\rho < 0)$: the proportion of replications in which the drop actually reduced true reliability. Table 1 summarises every estimand and its role.

Table 1: Summary of the estimands. All quantities in rows 2–3 are computed per strategy; for the conditional strategy they are additionally reported conditional on a drop having occurred.

#	Quantity	Definition	Role
1a	Calibration bias (full pool)	$\mathbb{E}[\hat{\alpha}(F) - \alpha_{\text{pop}}(F)]$	pipeline check; baseline for 2a
1b	Structural gap (full pool)	$\mathbb{E}[\alpha_{\text{pop}}(F) - \rho(F)]$	≤ 0; exists with no selection
1c	Total full-pool bias	$\mathbb{E}[\hat{\alpha}(F) - \rho(F)]$	1a + 1b; no-dropping benchmark
2a	Reported-alpha bias, alpha scale	$\mathbb{E}[\hat{\alpha}(S_s) - \alpha_{\text{pop}}(S_s)]$	statistical bias incl. winner's curse
2b	Reported-alpha bias, reliability scale	$\mathbb{E}[\hat{\alpha}(S_s) - \rho(S_s)]$	total bias in the reported estimate
3a	Observed gain	$\mathbb{E}[\widehat{\Delta}_{\text{obs}}]$	the trigger; researcher's-eye view
3b	Population alpha change	$\mathbb{E}[\Delta_{\text{pop}}]$	did the drop improve alpha in truth
3c	True reliability change	$\mathbb{E}[\Delta_\rho]$	did the drop improve measurement
3d	Selection optimism	$\mathbb{E}[\widehat{\Delta}_{\text{obs}} - \Delta_{\text{pop}}]$	winner's curse, paired per replicate
3e	P(drop)	$\Pr(\hat{\alpha}(-J^*) > \hat{\alpha}(F))$	conditional strategy only (forced $\equiv 1$)

#	Quantity	Definition	Role
3f	Harm rate	$\Pr(\Delta_\rho < 0)$	how often dropping reduced reliability
3g	Harm magnitude	$\mathbb{E}[\Delta_\rho \mid \Delta_\rho < 0]$	how large the harm is when it occurs
4	Prediction interval	$[\hat{q}_{2.5}(d),\ \hat{q}_{97.5}(d)]$ for d in 2a–3d	spread, not precision: one study's range

2.8. Performance measures and Monte Carlo uncertainty

Four types of performance measure were computed per condition over its replications.

1. Calibration and structural gap (full pool, no dropping): the mean signed difference between sample α and population α, which checks the pipeline is calibrated before any selection, and the mean difference between population α and ω, which is the baseline structural gap that exists regardless of dropping.
2. Bias in the reported α of the retained items, per strategy, against both population α and true reliability, reported both marginally and, for the conditional strategy, conditional on a drop having occurred.
3. Impact of item dropping: the expected change from full pool to retained items on all three scales, the selection-optimism gap, the probability that the conditional strategy drops at all, and the harm rate.
4. 95% prediction intervals (empirical 2.5th and 97.5th percentiles across replications) for every quantity in 2 and 3, reported alongside their means, describing the spread an individual study can encounter, which the mean alone does not.

All continuous estimands are means of a per-replicate difference. Writing d_k for the relevant difference in replication k of a condition, for example $\hat{\alpha}(S_s) - \rho(S_s)$ for the headline reliability-bias estimand,

$$\hat{\mathbb{E}}[d] = \frac{1}{K}\sum_{k=1}^{K} d_k, \qquad \widehat{\mathrm{MCSE}}(\hat{\mathbb{E}}[d]) = \frac{S_d}{\sqrt{K}}$$

where S_d is the standard deviation of d_k across the K replications. In the compendium these are computed with `simhelpers::calc_absolute()` (Joshi & Pustejovsky, 2022), which implements the Siepe et al. (2024) bias and Monte Carlo standard error formulas. That function requires a single fixed true parameter per condition, whereas here every replication draws its own loading vector and therefore has its own target. We therefore pass the per-replicate difference d_k itself as the estimate with a true parameter of 0, which is algebraically identical to bias against a varying target. Proportions take the binomial Monte Carlo standard error, $\widehat{\mathrm{MCSE}}(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/K}$.

For the conditional strategy, all measure-2 and measure-3 quantities are additionally summarised over only the replications in which a drop occurred: the number of such replications, K_{drop}, is itself random and is reported alongside every conditional-on-drop estimate. For measure 4 we report the empirical 2.5th and 97.5th percentiles of d_k,

$$\widehat{\mathrm{PI}}_{95}(d) = [\hat{q}_{2.5}(d_k),\ \hat{q}_{97.5}(d_k)]$$

The prediction-interval bounds are themselves Monte Carlo estimates and carry their own uncertainty, which cannot be assumed negligible: unlike a mean, a sample quantile's standard error depends on the density of the distribution at that quantile, $\mathrm{SE}(\hat{q}_p) \approx \sqrt{p(1-p)/K} \,/\, f(q_p)$, so it can be large wherever the tail is thin, and quantile estimates in heavy-tailed or sparse regions can be unstable. We therefore estimated a Monte Carlo standard error for every reported bound using the exact distribution-free order-statistic method (the 95% confidence interval for a population quantile lies between the order statistics whose ranks are the Binomial(K, p) percentiles), cross-checked against a nonparametric bootstrap, with which it agrees to within roughly 30% in every cell. The bounds turn out to be stable, as across all 300 cell-by-quantity combinations, the larger of the two bound standard errors is at most 2.2% of its interval's width (median 1.2%), so every interval in Table 7 and Figure 5 is accurate to well within the precision at which it is printed. As expected, quantiles are estimated less precisely than means, by a factor of up to 8 in the same cell, and the least stable bounds are the upper tails of the observed-gain and optimism distributions at the smallest sample size, which are the heaviest tails in the design. One degenerate case is worth noting: for the conditional strategy the lower bound of the observed gain sits on the point mass at zero contributed by the no-drop replications, so that bound has a Monte Carlo standard error of exactly zero, an artefact of the mixture rather than evidence of precision, and is another reason the conditional strategy's interval is more informative conditional on a drop having occurred.

Monte Carlo standard errors and prediction intervals are different quantities and should not be confused. An MCSE is $S_d/\sqrt{K}$ and describes how precisely the mean is estimated, decreasing as the number of iterations (K) grows. A prediction interval is driven by S_d itself and describes how much individual studies differ; as K grows its *estimate* stabilises but its *width* does not shrink, because the width is a property of the distribution rather than of how precisely we have sampled it. At K = 10,000 the MCSE is roughly one hundredth of the standard deviation, which is precisely why Aim 4 requires its own summaries rather than wider error bars on the existing ones.

Because each replication draws both a fresh item pool and a fresh sample from it, the prediction intervals marginalise over two sources of variation at once. They therefore describe the range across studies of psychological trait scales in general, which is the right quantity for a field-level claim, and not the range available to one researcher holding one fixed scale, which would require a nested design. Simulations were parallelised with reproducible L'Ecuyer-CMRG streams, so results are identical regardless of the number of workers used.

Table 2: Full item pool, no dropping. Estimate (+/-1 Monte Carlo standard error) of the calibration bias of sample alpha against population alpha (1a), the structural gap between population alpha and true reliability (1b), and their sum (1c), by item-pool size (k) and sample size (N).

k	N	1a. Calibration bias	1b. Structural gap	1c. Total bias
5	20	-0.0423 (0.0018)	-0.0242 (0.0002)	-0.0665 (0.0018)
	30	-0.0293 (0.0014)	-0.0244 (0.0002)	-0.0537 (0.0014)
	50	-0.0167 (0.0010)	-0.0240 (0.0002)	-0.0407 (0.0010)
	200	-0.0041 (0.0005)	-0.0240 (0.0002)	-0.0282 (0.0005)
	1000	-0.0008 (0.0002)	-0.0240 (0.0002)	-0.0248 (0.0003)

(continued)

k	N	1a. Calibration bias	1b. Structural gap	1c. Total bias
10	20	-0.0262 (0.0010)	-0.0144 (0.0001)	-0.0406 (0.0010)
	30	-0.0178 (0.0008)	-0.0144 (0.0001)	-0.0322 (0.0008)
	50	-0.0090 (0.0006)	-0.0142 (0.0001)	-0.0232 (0.0006)
	200	-0.0028 (0.0003)	-0.0143 (0.0001)	-0.0171 (0.0003)
	1000	-0.0004 (0.0001)	-0.0145 (0.0001)	-0.0149 (0.0001)
15	20	-0.0189 (0.0007)	-0.0102 (0.0001)	-0.0292 (0.0007)
	30	-0.0128 (0.0005)	-0.0102 (0.0001)	-0.0230 (0.0005)
	50	-0.0070 (0.0004)	-0.0103 (0.0001)	-0.0173 (0.0004)
	200	-0.0015 (0.0002)	-0.0102 (0.0001)	-0.0117 (0.0002)
	1000	-0.0003 (0.0001)	-0.0103 (0.0001)	-0.0106 (0.0001)
20	20	-0.0145 (0.0005)	-0.0079 (0.0000)	-0.0225 (0.0005)
	30	-0.0090 (0.0004)	-0.0079 (0.0000)	-0.0169 (0.0004)
	50	-0.0050 (0.0003)	-0.0079 (0.0000)	-0.0129 (0.0003)
	200	-0.0013 (0.0001)	-0.0079 (0.0000)	-0.0091 (0.0001)
	1000	-0.0002 (0.0001)	-0.0079 (0.0000)	-0.0081 (0.0001)
40	20	-0.0074 (0.0003)	-0.0041 (0.0000)	-0.0115 (0.0003)
	30	-0.0045 (0.0002)	-0.0041 (0.0000)	-0.0086 (0.0002)
	50	-0.0025 (0.0001)	-0.0041 (0.0000)	-0.0067 (0.0001)
	200	-0.0007 (0.0001)	-0.0041 (0.0000)	-0.0048 (0.0001)
	1000	-0.0001 (0.0000)	-0.0041 (0.0000)	-0.0042 (0.0000)

3. Results

Results are organised around the four performance measures. Every table reports all 25 conditions of the design rather than a selected or pooled subset, because the estimands vary substantially and non-monotonically across the grid and an average over the grid is an average over the design chosen here rather than over any population of real studies.

3.1. Calibration and the structural gap

Estimating the full pool Cronbach's α shows a small downward calibration bias at small samples (Table 2, column 1a; Figure 1, panel 1a). With small samples, the item covariances that α is built from are estimated with high sampling noise. The bias decreases toward zero as N grows at every item pool size, from -0.0423 at $k = 5$, $N = 20$ to -0.0008 at $N = 1000$, and is smaller for larger item pools even at fixed N (-0.0074 at $k = 40$, $N = 20$).

The structural gap between population α and true reliability behaves quite differently. It is a property of the loadings alone and does not depend on the sample at all, so its estimate is essentially invariant to N: within a fixed item pool size it varies by no more than 0.0004 across the full $N = 20$ to 1000 range, a residual difference attributable to Monte Carlo error in the loadings drawn rather than to sample size. The gap does shrink as pool size increases, from -0.0241 at $k = 5$ to -0.0041 at $k = 40$, because both population α and true reliability approach a ceiling near 1 as scales lengthen, compressing the distance between them even though the underlying inequality among item loadings is unchanged.

These two components sum to the total full pool bias (column 1c), which is negative everywhere: with no item dropping at all, the reported α understates the true reliability of the scale in every condition examined, by between 0.0042 and 0.0665.

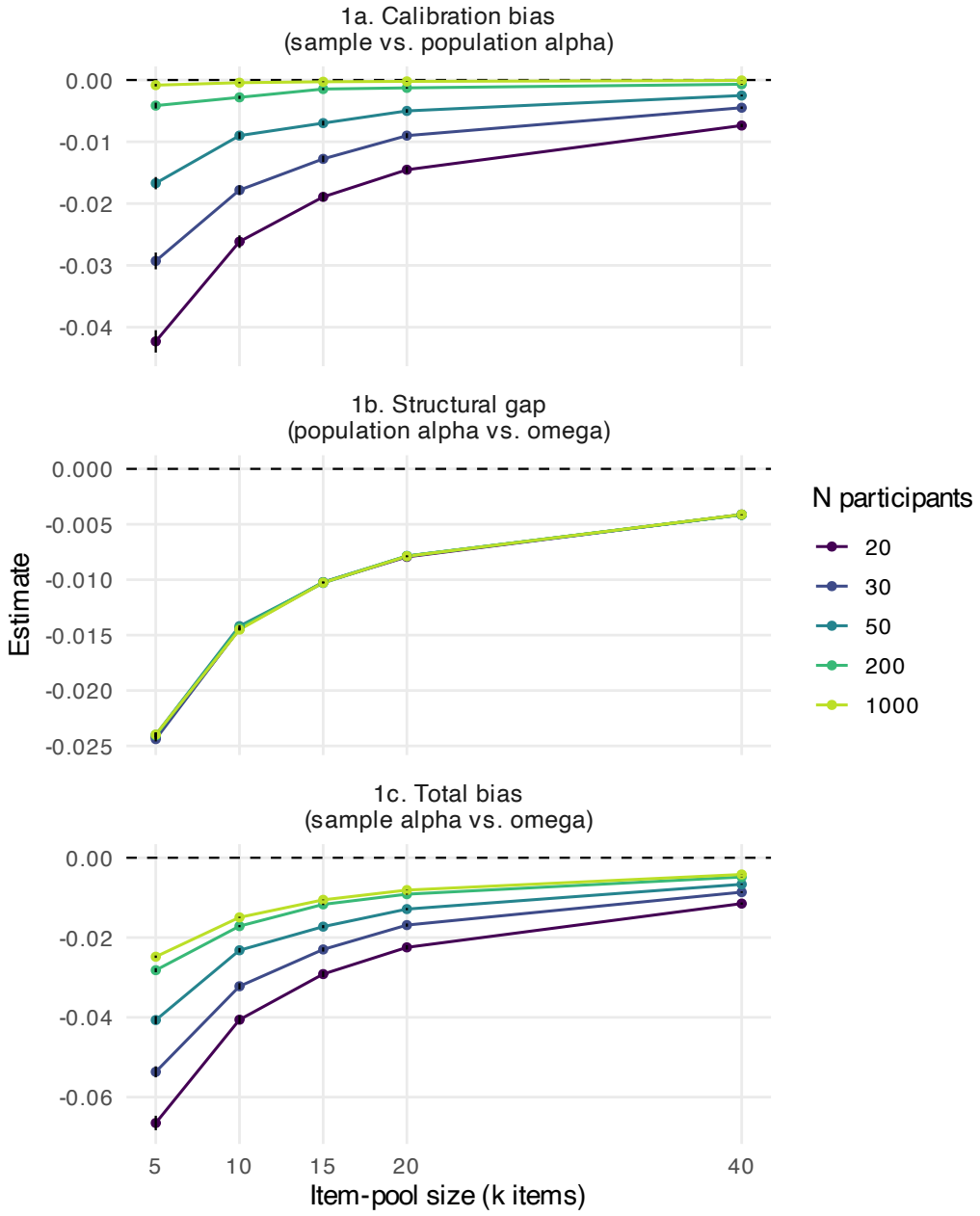


Figure 1: Full-pool calibration bias (1a), structural gap between population alpha and omega (1b), and their sum, the total bias against true reliability with no item dropping (1c), by item-pool size (x axis) and sample size (colour). Error bars are +/-1 Monte Carlo standard error and are narrower than the plotting symbol in most cells. Note the differing y-axis scales. The structural gap (1b) depends only on the loadings, so its five sample-size series coincide.

Table 3: Bias (+/-1 Monte Carlo standard error) of the reported (sample) alpha of the retained items against the true population alpha of those same items (2a) and against their true reliability (2b), by strategy, item-pool size (k), and sample size (N). The forced strategy drops in every replication by definition. Conditional-strategy columns are marginal, so replications in which nothing was dropped contribute the full-pool values.

		Forced		Conditional (marginal)	
k	N	2a vs. pop. alpha	2b vs. reliability	2a vs. pop. alpha	2b vs. reliability
5	20	0.0435 (0.0013)	0.0239 (0.0013)	0.0415 (0.0013)	0.0220 (0.0012)
	30	0.0315 (0.0011)	0.0131 (0.0010)	0.0293 (0.0010)	0.0110 (0.0010)
	50	0.0228 (0.0008)	0.0059 (0.0008)	0.0211 (0.0008)	0.0042 (0.0008)
	200	0.0064 (0.0004)	-0.0084 (0.0004)	0.0057 (0.0004)	-0.0091 (0.0004)
	1000	0.0015 (0.0002)	-0.0123 (0.0002)	0.0012 (0.0002)	-0.0125 (0.0002)
10	20	0.0057 (0.0008)	-0.0062 (0.0008)	0.0056 (0.0008)	-0.0063 (0.0008)
	30	0.0045 (0.0006)	-0.0069 (0.0006)	0.0044 (0.0006)	-0.0069 (0.0006)
	50	0.0048 (0.0005)	-0.0058 (0.0005)	0.0048 (0.0005)	-0.0059 (0.0005)
	200	0.0009 (0.0002)	-0.0090 (0.0002)	0.0008 (0.0002)	-0.0090 (0.0002)
	1000	0.0003 (0.0001)	-0.0093 (0.0001)	0.0003 (0.0001)	-0.0093 (0.0001)
15	20	-0.0017 (0.0006)	-0.0103 (0.0006)	-0.0017 (0.0006)	-0.0103 (0.0006)
	30	-0.0008 (0.0004)	-0.0092 (0.0004)	-0.0008 (0.0004)	-0.0092 (0.0004)
	50	0.0003 (0.0003)	-0.0078 (0.0003)	0.0003 (0.0003)	-0.0078 (0.0003)
	200	0.0005 (0.0002)	-0.0071 (0.0002)	0.0005 (0.0002)	-0.0071 (0.0002)
	1000	0.0001 (0.0001)	-0.0073 (0.0001)	0.0001 (0.0001)	-0.0073 (0.0001)
20	20	-0.0035 (0.0005)	-0.0103 (0.0005)	-0.0035 (0.0005)	-0.0103 (0.0005)
	30	-0.0015 (0.0003)	-0.0081 (0.0003)	-0.0015 (0.0003)	-0.0081 (0.0003)
	50	-0.0004 (0.0003)	-0.0068 (0.0003)	-0.0004 (0.0003)	-0.0068 (0.0003)
	200	-0.0000 (0.0001)	-0.0061 (0.0001)	-0.0000 (0.0001)	-0.0061 (0.0001)
	1000	0.0000 (0.0001)	-0.0059 (0.0001)	0.0000 (0.0001)	-0.0059 (0.0001)
40	20	-0.0039 (0.0002)	-0.0076 (0.0002)	-0.0039 (0.0002)	-0.0076 (0.0002)
	30	-0.0021 (0.0002)	-0.0058 (0.0002)	-0.0021 (0.0002)	-0.0058 (0.0002)
	50	-0.0010 (0.0001)	-0.0046 (0.0001)	-0.0010 (0.0001)	-0.0046 (0.0001)
	200	-0.0003 (0.0001)	-0.0038 (0.0001)	-0.0003 (0.0001)	-0.0038 (0.0001)
	1000	0.0000 (0.0000)	-0.0034 (0.0000)	0.0000 (0.0000)	-0.0034 (0.0000)

3.2. Bias in the reported alpha after item dropping

Against the population α of the retained items (Table 3, column 2a), the selection-induced optimism predicted by Kopalle & Lehmann (1997) is clearly present at the shortest scales, where the reported α overstates the population α of the very items it describes by 0.0435 at $k = 5$, $N = 20$. It shrinks rapidly with both factors, and at $k \geq 15$ with small samples it turns slightly negative, because the small-sample downward calibration bias of α documented above is then larger than the optimism the argmax can extract from a pool that is already long enough for α to be estimated fairly stably. Selection optimism is therefore not uniformly larger in longer scales, contrary to the intuition that more candidate items means more opportunity to capitalise on noise. With k items there are more candidates, but each contributes proportionally less to α, and the second effect dominates.

Against true reliability (column 2b), which is the applied target because α is reported as a reliability estimate, the two components combine. The reported α overstates true reliability only in the shortest scales at small samples, in 3 of the 25 conditions, all at $k = 5$: by 0.0239 at $N = 20$, falling to 0.0059 at $N = 50$. In every other condition it

understates true reliability, by between 0.003 and 0.012. The sign of the bias in a reported α after item dropping is therefore not fixed: it depends on where a given study sits relative to selection optimism on one side and small-sample calibration bias plus the structural gap on the other.

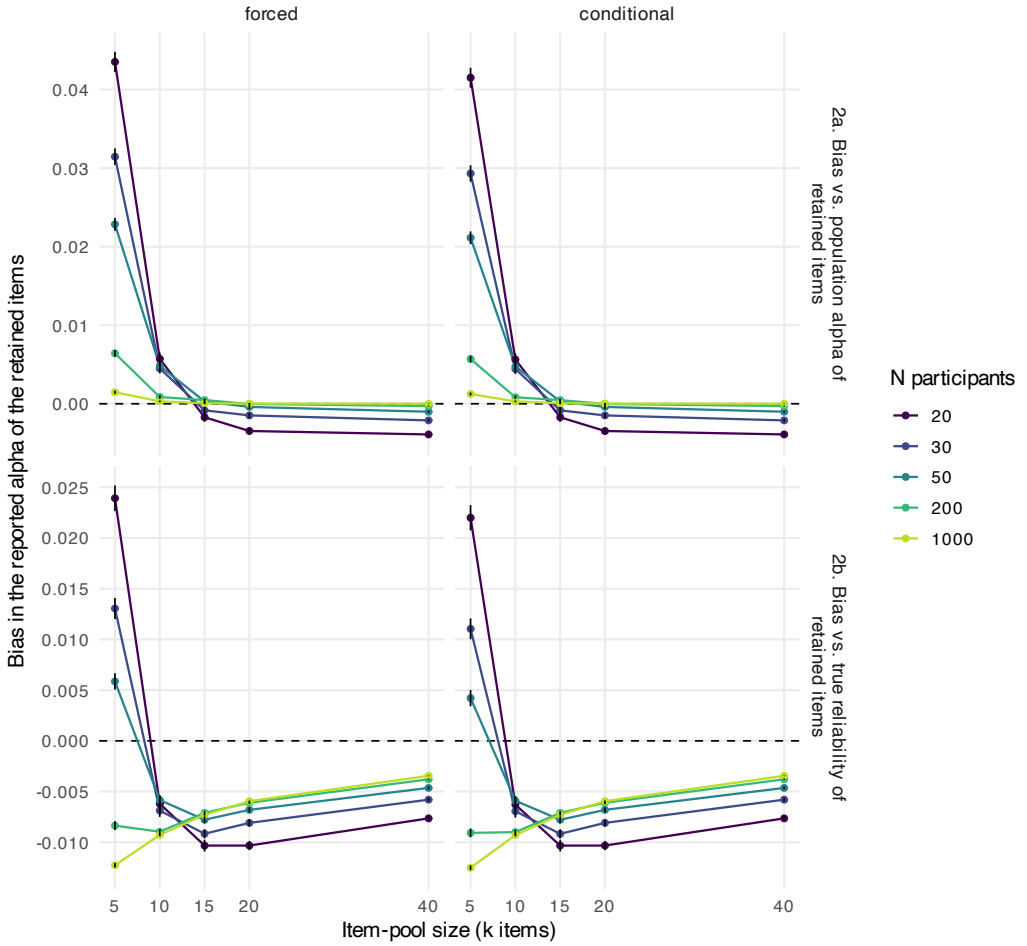


Figure 2: Bias in the reported alpha of the retained items against the population alpha of those items (top row, 2a) and against their true reliability (bottom row, 2b), for the forced (left) and conditional (right) strategies, by item-pool size and sample size. Error bars are +/-1 Monte Carlo standard error. The two strategies are near-identical in every panel, consistent with their shared nomination rule. Note the differing y-axis scales between rows.

The same two estimands can be shown on the scale an applied reader thinks in, which is also the form in which Kopalle & Lehmann (1997) summarised their own results and is therefore the directly comparable presentation: the true value of the retained items on the horizontal axis, the reported sample α of those same items on the vertical axis, with the identity line as the reference (Figure 3). Vertical distance above the identity line is exactly the bias of estimand 2a (top row) or 2b (bottom row), and a no-dropping series is

included so that the effect of selection is visible against the baseline of not selecting at all. Because every replication draws its own loadings, the true value varies continuously within a condition rather than being fixed, so replications are binned by it at a width of 0.05 and the mean reported α is plotted per bin. Bins containing fewer than 20 replications are not plotted, which removes 0.05% of replications in the sparse tails of the true-value distribution.

Two features are worth reading off it. First, the no-dropping series does not lie on the identity line: against the population α of the retained items (top row) it sits below it, by 0.010 on average where the true value exceeds 0.45 and by 0.025 where the true value is lower. This is the small-sample calibration bias seen from a different angle, and it is largest at low true values because those are the short, weakly loaded pools whose covariances are estimated least precisely. Any assessment of what selection does has to be read against that sloping baseline rather than against the identity line.

Second, both dropping series sit above the no-dropping series everywhere, and the vertical distance between them is the selection effect in its purest form, since the three series differ only in whether and how an item was chosen. At high true values that shift roughly cancels the calibration bias, leaving the reported α close to correct (0.001 for the forced rule). At low true values it substantially overshoots, leaving the reported α 0.070 above the truth. This is the Kopalle and Lehmann result in its original form, and it makes clear that their "alpha inflation" is not a uniform upward shift but something that bites hardest exactly where reliability is already poor.

The bottom row is the same replications compared against true reliability instead. Every series moves down by the structural gap, which is why the no-dropping series now sits 0.072 below the identity line at low true values. Dropping partly offsets that displacement rather than compounding it, which is the graphical form of the sign reversal in Table 3: at high true values the forced rule lands 0.007 below the truth, and only at low true values does it end up above it (0.041).

These curves must not be read backwards. They give the expected reported α given the truth. The question an applied researcher faces is the reverse one, since they know their reported α and want the truth, but the expected true value given a reported α is a different quantity that these curves do not provide, as it depends also on how often each true value arises in the first place. Reading horizontally from a reported α across to a curve would overstate what can be inferred from it.

Table 4: Conditional strategy, restricted to the replications in which a drop actually occurred. Estimate (+/-1 Monte Carlo standard error) of the bias in the reported alpha against population alpha (2a) and against true reliability (2b), the mean true reliability change (3c), and the harm rate among these drops (3f; percent). K(drop) is the realised number of replications in which a drop occurred, out of 10,000. The given-drop harm rate has a different denominator from the marginal harm rate in the impact table below, which counts no-drop replications as non-harm.

k	N	K(drop)	2a vs. pop. alpha	2b vs. reliability	3c. Reliability change	3f. Harm given drop
5	20	9,102	0.0391 (0.0014)	0.0190 (0.0013)	0.0023 (0.0005)	46.7 (0.52)
	30	8,868	0.0282 (0.0011)	0.0091 (0.0011)	0.0102 (0.0005)	39.4 (0.52)
	50	8,560	0.0209 (0.0009)	0.0030 (0.0009)	0.0167 (0.0004)	33.1 (0.51)
	200	8,013	0.0054 (0.0005)	-0.0109 (0.0005)	0.0296 (0.0004)	16.0 (0.41)
	1000	7,794	0.0013 (0.0002)	-0.0143 (0.0003)	0.0339 (0.0003)	7.2 (0.29)

(continued)

k	N	K(drop)	2a vs. pop. alpha	2b vs. reliability	3c. Reliability change	3f. Harm given drop
10	20	9,874	0.0052 (0.0008)	-0.0067 (0.0008)	0.0067 (0.0002)	32.0 (0.47)
	30	9,838	0.0041 (0.0006)	-0.0074 (0.0006)	0.0089 (0.0002)	26.9 (0.45)
	50	9,750	0.0045 (0.0005)	-0.0063 (0.0005)	0.0110 (0.0001)	19.8 (0.40)
	200	9,641	0.0007 (0.0002)	-0.0093 (0.0002)	0.0146 (0.0001)	7.6 (0.27)
	1000	9,614	0.0003 (0.0001)	-0.0095 (0.0001)	0.0159 (0.0001)	2.5 (0.16)
15	20	9,982	-0.0017 (0.0006)	-0.0104 (0.0006)	0.0051 (0.0001)	25.0 (0.43)
	30	9,978	-0.0009 (0.0004)	-0.0092 (0.0004)	0.0062 (0.0001)	19.0 (0.39)
	50	9,970	0.0003 (0.0003)	-0.0078 (0.0003)	0.0073 (0.0001)	12.8 (0.33)
	200	9,939	0.0005 (0.0002)	-0.0071 (0.0002)	0.0088 (0.0001)	3.7 (0.19)
	1000	9,919	0.0000 (0.0001)	-0.0073 (0.0001)	0.0096 (0.0001)	0.8 (0.09)
20	20	9,999	-0.0035 (0.0005)	-0.0103 (0.0005)	0.0036 (0.0000)	20.4 (0.40)
	30	9,998	-0.0015 (0.0003)	-0.0081 (0.0003)	0.0043 (0.0000)	15.2 (0.36)
	50	9,993	-0.0004 (0.0003)	-0.0068 (0.0003)	0.0050 (0.0000)	8.9 (0.28)
	200	9,989	-0.0000 (0.0001)	-0.0061 (0.0001)	0.0059 (0.0000)	1.8 (0.13)
	1000	9,985	0.0000 (0.0001)	-0.0060 (0.0001)	0.0063 (0.0000)	0.3 (0.06)
40	20	10,000	-0.0039 (0.0002)	-0.0076 (0.0002)	0.0014 (0.0000)	12.0 (0.33)
	30	10,000	-0.0021 (0.0002)	-0.0058 (0.0002)	0.0016 (0.0000)	7.0 (0.26)
	50	10,000	-0.0010 (0.0001)	-0.0046 (0.0001)	0.0018 (0.0000)	3.2 (0.18)
	200	10,000	-0.0003 (0.0001)	-0.0038 (0.0001)	0.0021 (0.0000)	0.1 (0.04)
	1000	10,000	0.0000 (0.0000)	-0.0034 (0.0000)	0.0022 (0.0000)	0.0 (0.00)

Restricting the conditional strategy to the replications in which it actually acted (Table 4) leaves this pattern almost unchanged, because both strategies remove the same nominated item whenever the conditional strategy drops at all. The realised K_{drop} shrinks furthest at the smallest pool size, where the conditional rule declines to act increasingly often as N grows, from 9.0% of replications at $N = 20$ to 22.1% at $N = 1000$. This is consistent with greater sampling precision at larger N leaving less noise for the argmax to exploit.

3.3. Impact of item dropping

Whether an item is dropped at all, and whether doing so helps or harms the scale, are both determined by the number of items and the sample size (Table 5, Figure 4). The propensity of the conditional strategy to drop an item is highest exactly where its judgment is least trustworthy. At the smallest pool size, Pr(drop) falls from 91.0% at $N = 20$ to 77.9% at $N = 1000$. At the largest pool size it remains at or near 100% regardless of sample size: with more items available there is almost always some single item whose removal raises the sample α, whether or not that increase reflects a real improvement to the scale.

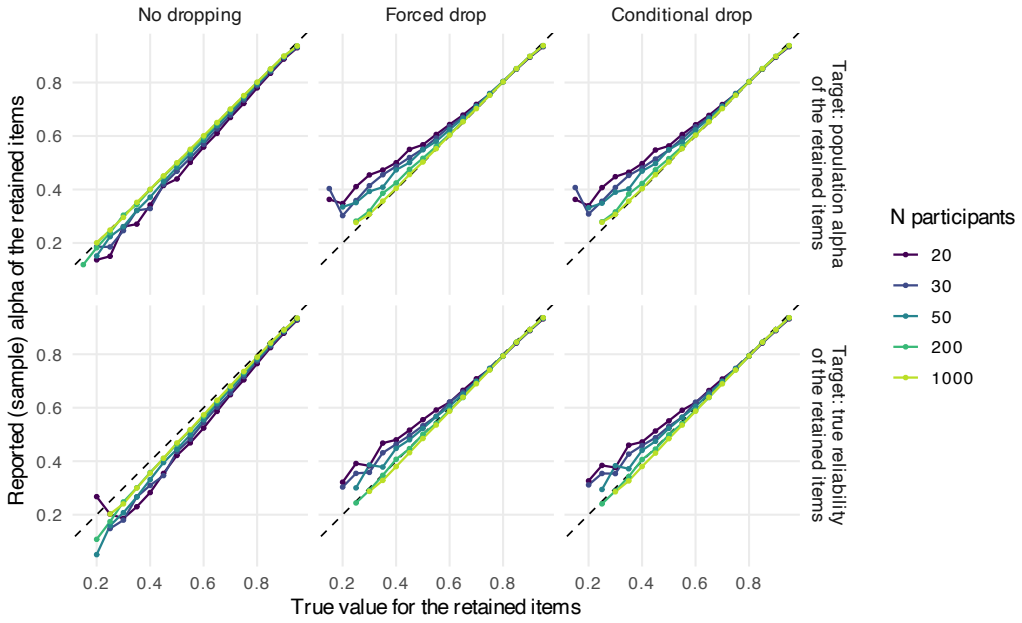


Figure 3: Calibration view: the reported (sample) alpha of the retained items against their true value, in the presentation used by Kopalle and Lehmann (1997). Columns are the two dropping strategies plus a no-dropping reference, rows are the two population targets. The dashed line is the identity: points on it mean the reported alpha is exactly right for the items retained, and points above it mean it overstates them. Replications are binned by their true value at a width of 0.05 and averaged within bin. Bins with fewer than 20 replications are omitted. Binning is presentational only and no estimand is defined in terms of it.

Table 5: Proportion of replications in which the conditional strategy dropped an item (3e; the forced strategy drops in all replications by definition), the proportion in which the drop reduced true reliability (3f, the harm rate), and the mean size of that reduction when it occurred (3g, harm magnitude; forced strategy), by item-pool size (k) and sample size (N). Proportions are percentages with +/-1 Monte Carlo standard error in percentage points; conditional-strategy harm is marginal (no-drop replications count as non-harm; see the conditional-on-drop table for the given-drop rate). The final column is the proportion of forced-strategy replications in which the apparent in-sample gain was not positive, i.e. in which the drop did not even appear to help. An em-dash marks the one cell with no harmful replications, where the conditional mean does not exist.

k	N	P(drop)	Harm, forced	Harm, cond.	3g. Mean harm	No gain, forced
5	20	91.0 (0.29)	49.6 (0.50)	42.5 (0.49)	-0.0398 (0.0005)	9.0
	30	88.7 (0.32)	43.9 (0.50)	34.9 (0.48)	-0.0348 (0.0005)	11.3
	50	85.6 (0.35)	39.3 (0.49)	28.3 (0.45)	-0.0290 (0.0004)	14.4
	200	80.1 (0.40)	28.9 (0.45)	12.8 (0.33)	-0.0182 (0.0003)	19.9
	1000	77.9 (0.41)	25.3 (0.43)	5.6 (0.23)	-0.0137 (0.0002)	22.1
10	20	98.7 (0.11)	32.7 (0.47)	31.6 (0.47)	-0.0107 (0.0002)	1.3
	30	98.4 (0.13)	27.6 (0.45)	26.5 (0.44)	-0.0087 (0.0002)	1.6
	50	97.5 (0.16)	21.1 (0.41)	19.4 (0.40)	-0.0072 (0.0001)	2.5
	200	96.4 (0.19)	10.0 (0.30)	7.4 (0.26)	-0.0040 (0.0001)	3.6
	1000	96.1 (0.19)	5.6 (0.23)	2.4 (0.15)	-0.0026 (0.0001)	3.9
15	20	99.8 (0.04)	25.1 (0.43)	24.9 (0.43)	-0.0046 (0.0001)	0.2
	30	99.8 (0.05)	19.1 (0.39)	19.0 (0.39)	-0.0037 (0.0001)	0.2
	50	99.7 (0.05)	13.0 (0.34)	12.8 (0.33)	-0.0031 (0.0001)	0.3
	200	99.4 (0.08)	4.1 (0.20)	3.6 (0.19)	-0.0015 (0.0001)	0.6
	1000	99.2 (0.09)	1.4 (0.12)	0.8 (0.09)	-0.0011 (0.0001)	0.8
20	20	100.0 (0.01)	20.4 (0.40)	20.4 (0.40)	-0.0025 (0.0000)	0.0
	30	100.0 (0.01)	15.2 (0.36)	15.2 (0.36)	-0.0021 (0.0000)	0.0
	50	99.9 (0.03)	8.9 (0.28)	8.9 (0.28)	-0.0016 (0.0000)	0.1
	200	99.9 (0.03)	1.9 (0.14)	1.8 (0.13)	-0.0008 (0.0001)	0.1
	1000	99.9 (0.04)	0.4 (0.07)	0.3 (0.06)	-0.0005 (0.0001)	0.1
40	20	100.0 (0.00)	12.0 (0.33)	12.0 (0.33)	-0.0006 (0.0000)	0.0
	30	100.0 (0.00)	7.0 (0.26)	7.0 (0.26)	-0.0005 (0.0000)	0.0
	50	100.0 (0.00)	3.2 (0.18)	3.2 (0.18)	-0.0004 (0.0000)	0.0
	200	100.0 (0.00)	0.1 (0.04)	0.1 (0.04)	-0.0002 (0.0000)	0.0
	1000	100.0 (0.00)	0.0 (0.00)	0.0 (0.00)	—	0.0

The consequences of removing the item are given by the harm rate, the proportion of replications in which the drop reduced true reliability. At the smallest scale and sample size the forced strategy's harm rate is 49.6%: the nominated item is very nearly as likely to be an artefact of that particular sample as it is to be a genuine weak point in the scale. Harm falls steeply as either factor increases, but it does not vanish with sample size alone. At $N = 1000$ the forced strategy still reduces true reliability in 25.3% of replications at $k = 5$, and the rate reaches 0.0% only in the largest scale and sample combination. Long scales and large samples are both needed – neither alone is sufficient.

The harm rate counts sign only, so it must be read alongside the harm *magnitude*, the mean reliability loss among the replications in which harm occurred (Table 5, column 3g). The two together show that the harm rates are not comparable across scale lengths without the magnitudes. At $k = 5$, $N = 20$, harm is both frequent and material: the mean loss given harm is -0.040, comparable in size to the mean *gain* among helpful drops. At $k = 40$ the harm rate of 12.0% at $N = 20$ counts losses averaging only -0.0006, trivial

beside those at short scales. The practical risk is therefore concentrated where the rate and the magnitude are large together, in short scales at small samples.

The conditional strategy's harm rates require care with denominators, because two are available and they answer different questions. Marginally, across all replications including those where nothing was dropped, the conditional rule harms in 42.5% of replications at $k = 5$, $N = 20$, versus 49.6% under the forced rule. But conditional on the rule actually firing, which is the risk a researcher who has just been told to drop an item faces, the harm rate is 46.7% (Table 4): barely below the forced rule's. Declining to act in the replications with the weakest apparent improvement lowers the incidence of harm across all studies mostly by gambling less often, not by gambling better. The gating rule is a filter on how often the gamble is taken, and only weakly on whether it pays off.

An important qualification concerns how the harm rate should be described. It is tempting to say the drop harms reliability "while appearing to improve α every time", but this holds only for the conditional strategy, for which $\widehat{\Delta}_{\text{obs}} > 0$ is true by construction whenever it acts. The forced strategy has no such guarantee: its apparent gain was not positive in 9.0% of replications at $k = 5$, $N = 20$, rising to 22.1% at $N = 1000$ (Table 5, final column). The strongest claim the design supports without qualification is therefore the conditional one: at $k = 5$, $N = 20$, the rule that acts only when the in-sample α improves nevertheless reduced true reliability in 46.7% of the replications in which it acted.

The same pattern appears in the magnitude of what item dropping does to the scale (Table 6). The observed sample gain $\widehat{\Delta}_{\text{obs}}$ is positive on average in every condition, including those with the highest harm rates. The population α change Δ_{pop} and the true reliability change Δ_{ρ} are both consistently smaller, and the gap between $\widehat{\Delta}_{\text{obs}}$ and Δ_{pop}, the selection optimism, is largest exactly where the harm rate is highest: 0.086 at $k = 5$, $N = 20$, falling to 0.0001 at $k = 40$, $N = 1000$. The conditions producing the most inflated apparent gain are the same conditions producing the highest probability that the underlying decision was actively counterproductive. Most strikingly, at $k = 5$, $N = 20$ the forced rule's mean apparent gain of 0.090 in sample α corresponds to a mean change in true reliability of -0.0007, which is to say essentially none at all: almost the entire apparent improvement is selection optimism.

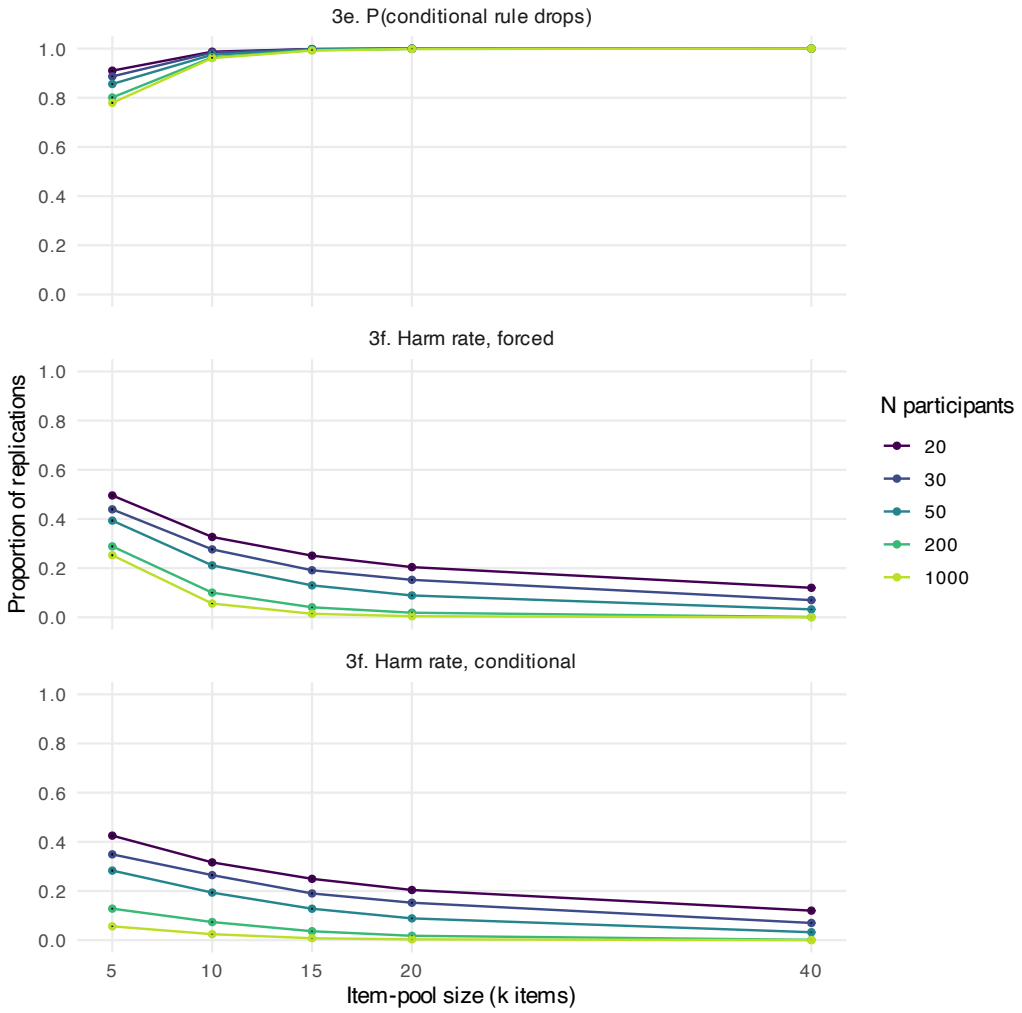


Figure 4: Probability that the conditional strategy drops an item (left) and the probability that the drop reduced true reliability, i.e. the harm rate, for each strategy (centre and right), by item-pool size and sample size. Error bars are +/-1 Monte Carlo standard error. The conditions in which the conditional rule is most likely to act are the conditions in which acting is most likely to harm the scale.

Table 6: Mean change (+/-1 Monte Carlo standard error) from full pool to retained items on three scales: observed sample alpha (3a), population alpha (3b), and true reliability (3c), together with the selection optimism (3d = 3a - 3b, computed pairwise within replicate), for the forced strategy, by item-pool size (k) and sample size (N). Conditional-strategy values are given in the compendium, they differ only through the replications in which no drop occurred, which contribute zeros.

k	N	3a. Observed	3b. Population alpha	3c. True reliability	3d. Optimism
5	20	0.0897 (0.0011)	0.0038 (0.0006)	-0.0007 (0.0005)	0.0858 (0.0010)
	30	0.0726 (0.0008)	0.0118 (0.0006)	0.0058 (0.0005)	0.0607 (0.0008)
	50	0.0576 (0.0007)	0.0180 (0.0006)	0.0110 (0.0004)	0.0395 (0.0006)
	200	0.0404 (0.0005)	0.0298 (0.0005)	0.0206 (0.0003)	0.0106 (0.0003)
	1000	0.0359 (0.0005)	0.0336 (0.0005)	0.0234 (0.0003)	0.0023 (0.0001)
10	20	0.0409 (0.0004)	0.0090 (0.0002)	0.0065 (0.0002)	0.0319 (0.0004)
	30	0.0340 (0.0003)	0.0116 (0.0002)	0.0086 (0.0002)	0.0223 (0.0003)
	50	0.0279 (0.0002)	0.0141 (0.0002)	0.0106 (0.0001)	0.0138 (0.0002)
	200	0.0221 (0.0002)	0.0184 (0.0002)	0.0139 (0.0001)	0.0037 (0.0001)
	1000	0.0209 (0.0002)	0.0201 (0.0002)	0.0152 (0.0001)	0.0007 (0.0000)
15	20	0.0239 (0.0002)	0.0067 (0.0001)	0.0051 (0.0001)	0.0172 (0.0002)
	30	0.0200 (0.0001)	0.0080 (0.0001)	0.0061 (0.0001)	0.0119 (0.0001)
	50	0.0167 (0.0001)	0.0095 (0.0001)	0.0072 (0.0001)	0.0073 (0.0001)
	200	0.0134 (0.0001)	0.0115 (0.0001)	0.0088 (0.0001)	0.0019 (0.0000)
	1000	0.0127 (0.0001)	0.0124 (0.0001)	0.0095 (0.0001)	0.0003 (0.0000)
20	20	0.0158 (0.0001)	0.0047 (0.0001)	0.0036 (0.0000)	0.0111 (0.0001)
	30	0.0131 (0.0001)	0.0056 (0.0001)	0.0043 (0.0000)	0.0075 (0.0001)
	50	0.0110 (0.0001)	0.0064 (0.0001)	0.0050 (0.0000)	0.0046 (0.0001)
	200	0.0090 (0.0001)	0.0077 (0.0000)	0.0059 (0.0000)	0.0012 (0.0000)
	1000	0.0085 (0.0000)	0.0082 (0.0000)	0.0063 (0.0000)	0.0002 (0.0000)
40	20	0.0052 (0.0000)	0.0017 (0.0000)	0.0014 (0.0000)	0.0035 (0.0000)
	30	0.0044 (0.0000)	0.0020 (0.0000)	0.0016 (0.0000)	0.0024 (0.0000)
	50	0.0038 (0.0000)	0.0023 (0.0000)	0.0018 (0.0000)	0.0015 (0.0000)
	200	0.0031 (0.0000)	0.0027 (0.0000)	0.0021 (0.0000)	0.0004 (0.0000)
	1000	0.0029 (0.0000)	0.0029 (0.0000)	0.0022 (0.0000)	0.0001 (0.0000)

Table 7: Mean [95% prediction interval] of each per-replication quantity, for the forced strategy, by item-pool size (k) and sample size (N). The interval is the empirical 2.5th to 97.5th percentile range across the 10,000 replications of a condition. It describes the spread across individual studies, not the precision of the mean, and unlike a Monte Carlo standard error it does not shrink as the number of replications grows.

k	N	3a. Observed gain	3c. True reliability change	2b. Bias vs. reliability
5	20	0.090 [-0.013, 0.360]	-0.001 [-0.107, 0.094]	0.024 [-0.267, 0.243]
	30	0.073 [-0.017, 0.289]	0.006 [-0.090, 0.097]	0.013 [-0.221, 0.199]
	50	0.058 [-0.019, 0.227]	0.011 [-0.074, 0.097]	0.006 [-0.173, 0.159]
	200	0.040 [-0.024, 0.166]	0.021 [-0.040, 0.099]	-0.008 [-0.103, 0.073]
	1000	0.036 [-0.025, 0.155]	0.023 [-0.028, 0.099]	-0.012 [-0.065, 0.027]
10	20	0.041 [0.002, 0.138]	0.006 [-0.025, 0.039]	-0.006 [-0.199, 0.111]
	30	0.034 [0.001, 0.110]	0.009 [-0.019, 0.040]	-0.007 [-0.155, 0.093]
	50	0.028 [0.000, 0.085]	0.011 [-0.015, 0.040]	-0.006 [-0.114, 0.073]
	200	0.022 [-0.001, 0.066]	0.014 [-0.006, 0.041]	-0.009 [-0.063, 0.034]
	1000	0.021 [-0.001, 0.062]	0.015 [-0.002, 0.043]	-0.009 [-0.036, 0.011]
15	20	0.024 [0.003, 0.075]	0.005 [-0.010, 0.021]	-0.010 [-0.156, 0.072]
	30	0.020 [0.003, 0.057]	0.006 [-0.007, 0.022]	-0.009 [-0.116, 0.061]
	50	0.017 [0.002, 0.046]	0.007 [-0.005, 0.023]	-0.008 [-0.088, 0.047]
	200	0.013 [0.002, 0.035]	0.009 [-0.001, 0.023]	-0.007 [-0.044, 0.022]
	1000	0.013 [0.001, 0.033]	0.009 [0.001, 0.023]	-0.007 [-0.025, 0.007]
20	20	0.016 [0.003, 0.047]	0.004 [-0.005, 0.013]	-0.010 [-0.122, 0.053]
	30	0.013 [0.003, 0.035]	0.004 [-0.004, 0.014]	-0.008 [-0.093, 0.044]
	50	0.011 [0.002, 0.028]	0.005 [-0.002, 0.014]	-0.007 [-0.066, 0.035]
	200	0.009 [0.002, 0.021]	0.006 [0.000, 0.014]	-0.006 [-0.034, 0.016]
	1000	0.008 [0.002, 0.020]	0.006 [0.001, 0.014]	-0.006 [-0.019, 0.005]
40	20	0.005 [0.001, 0.014]	0.001 [-0.001, 0.004]	-0.008 [-0.070, 0.025]
	30	0.004 [0.001, 0.010]	0.002 [-0.001, 0.004]	-0.006 [-0.051, 0.021]
	50	0.004 [0.001, 0.008]	0.002 [-0.000, 0.004]	-0.005 [-0.036, 0.017]
	200	0.003 [0.001, 0.006]	0.002 [0.001, 0.004]	-0.004 [-0.018, 0.008]
	1000	0.003 [0.001, 0.006]	0.002 [0.001, 0.004]	-0.003 [-0.010, 0.002]

3.4. What a single study can encounter

Every quantity above is a long-run average over 10,000 replications. The prediction intervals in Table 7 and Figure 5 describe the spread instead, which is to say the range of outcomes that a single study, run once using this rule, can encounter.

The intervals narrow sharply as either design factor grows. For the bias in the reported α against true reliability, the interval spans 0.510 at $k = 5$, $N = 20$ ([-0.267, 0.243]) and only 0.012 at $k = 40$, $N = 1000$ ([-0.010, 0.002]). The true reliability change narrows correspondingly, from 0.200 to 0.003 wide.

The gap between these intervals and the corresponding Monte Carlo standard errors is the substantive point of this aim. At $k = 5$, $N = 20$, the mean bias against true reliability is 0.024 with a Monte Carlo standard error of 0.0013, so the long-run average is pinned down to within a thousandth. Yet the individual studies making up that average span [-0.267, 0.243], a range roughly 404 times wider than the standard error.

That ratio is not a finding about item dropping, it is close to an identity. For any approximately normal quantity the interval width is about $3.92S_d$ while the standard error is $S_d/\sqrt{K}$, so their ratio is about $3.92\sqrt{K} \approx 392$ at $K = 10{,}000$ regardless of the

condition, and indeed it varies only between 345 and 424 across the entire design. Running more replications therefore raises this ratio rather than lowering it. The precision of the reported mean and the spread that an individual study faces are not merely different in size, they move in opposite directions as a simulation is scaled up. A researcher reporting an α after dropping an item from a five-item scale measured on twenty participants is drawing from the wide distribution, not from a neighbourhood of the mean, and no amount of additional simulation narrows what they face.

3.5. Post hoc analysis: can a stricter decision rule fix the decision?

The conditional strategy's gate of any apparent improvement at all is the most permissive rule available. An obvious practical question is whether a stricter gate rescues the practice, such that an item is dropped only if the apparent gain exceeds some threshold t. Because the simulation cache records the full-pool and dropped-pool sample alphas and both population targets for every replication, any such rule is computable from the same 250,000 replications without new simulation, with $t = 0$ reproducing the conditional strategy exactly. This analysis was not part of the planned design and should be read as post hoc. Table 8 reports, for $t \in \{0, 0.01, 0.02, 0.05\}$, how often each rule fires and the probability that firing harmed true reliability.

Table 8: Post hoc thresholded decision rules: drop only if the apparent in-sample gain exceeds t. For each rule and condition, the percentage of replications in which the rule fires, and the percentage of those firings that reduced true reliability (harm given fire). t = 0 is the conditional strategy. All values are percentages; the binomial Monte Carlo standard error is at most 0.5 percentage points for fire rates, and for harm rates it is computed on the realised number of firings. An em-dash marks cells in which the rule never fired, where harm given fire is undefined.

		Fires (% of replications)				Harm given fire (%)			
k	N	t = 0	t = .01	t = .02	t = .05	t = 0	t = .01	t = .02	t = .05
5	20	91.0	84.1	76.5	55.7	46.7	44.8	42.9	38.2
	30	88.7	80.9	72.9	50.5	39.4	36.7	34.2	28.6
	50	85.6	76.4	67.1	43.7	33.1	29.3	25.7	18.2
	200	80.1	69.7	59.4	33.6	16.0	11.2	7.0	1.8
	1000	77.9	66.9	56.0	30.0	7.2	2.2	0.6	0.0
10	20	98.7	86.6	68.4	28.0	32.0	28.9	25.4	19.2
	30	98.4	84.0	62.9	20.6	26.9	22.4	18.0	11.3
	50	97.5	80.2	56.3	13.9	19.8	14.0	9.0	4.1
	200	96.4	73.9	46.1	7.4	7.6	2.0	0.3	0.0
	1000	96.1	72.5	43.5	5.6	2.5	0.0	0.0	0.0
15	20	99.8	78.6	46.6	8.3	25.0	20.4	16.1	13.7
	30	99.8	74.8	39.0	4.2	19.0	13.7	8.7	5.3
	50	99.7	69.0	29.6	1.6	12.8	6.7	3.3	0.6
	200	99.4	59.0	18.9	0.3	3.7	0.3	0.1	0.0
	1000	99.2	57.0	16.3	0.2	0.8	0.0	0.0	0.0
20	20	100.0	63.8	25.0	2.1	20.4	15.4	13.1	14.6
	30	100.0	56.1	17.0	0.5	15.2	8.5	5.4	2.0
	50	99.9	47.6	9.6	0.1	8.9	3.1	1.6	0.0
	200	99.9	34.6	3.4	0.0	1.8	0.0	0.0	—
	1000	99.9	31.7	2.6	0.0	0.3	0.0	0.0	—
40	20	100.0	7.3	0.4	0.0	12.0	10.0	17.1	0.0
	30	100.0	2.7	0.1	0.0	7.0	4.0	0.0	—
	50	100.0	0.7	0.0	0.0	3.2	0.0	—	—
	200	100.0	0.1	0.0	0.0	0.1	0.0	—	—
	1000	100.0	0.0	0.0	0.0	0.0	—	—	—

The pattern splits cleanly on sample size. At $N \geq 200$, thresholds work to maintain reliability: requiring an apparent gain of at least .02 keeps the harm-given-fire probability at or below 7.0% at $N = 200$ wherever the rule fires at all (7.0% at $k = 5$, 0.3% at $k = 10$), and near zero at $N = 1000$. At the longest scales the .02 gate simply never fires, because a single item out of 40 cannot move α by that much; abstention of dropping an item is the correct behaviour there, since those are also the cells where dropping has the least to offer. At small samples, thresholds fail: at $k = 5$, $N = 20$, even demanding an apparent gain of .05, which more than halves how often the rule fires, still leaves a 38.2% chance that the drop harms true reliability, and at $N = 50$ the strictest gate examined still carries 18.2%. The reason is the one documented under Aim 4: at small N the apparent gain is dominated by selection over noise, so conditioning on a *larger* apparent gain selects for more extreme noise nearly as often as for genuinely worse items, and no threshold on an in-sample quantity can separate the two.

As a rule of thumb, an alpha-if-item-removed recommendation should not be acted on at all in small samples ($N \leq 50$), whatever the apparent improvement; in samples of

several hundred or more, acting only when the apparent gain exceeds .02 keeps the risk of harming the scale below roughly one in fourteen even for five-item scales, and negligible for longer ones.

4. Discussion

Dropping items to improve Cronbach's α involves two competing pressures: shorter scales produce lower α, but removing poorly performing items raises it. The aggregate impact of these pressures on the reliability of the reported α was not previously known, nor was the impact of conditioning item dropping decisions on the same data used to report α. Our results show that the practice is more consequential, and in more directions at once, than prior work suggested.These results demonstrate that item dropping can limit interpretability of a study's reported α estimates. We provide four specific findings using real-world psychological scale responses as a basis for simulation. First, α has two independent reasons to be untrustworthy before any item is dropped at all: ordinary small-sample noise, which fades with N (from -0.042 at $N = 20$ to -0.0008 at $N = 1000$ for $k = 5$), and a structural ceiling on what α can measure relative to true reliability, which does not fade with N at all, only with k (from -0.024 at $k = 5$ to -0.004 at $k = 40$). Second, introducing the selection step adds a third source of distortion that interacts with the first two and flips the overall sign of the bias depending on where in the design a study sits. Third, the bias in the number is almost beside the point: the decision itself is frequently inadvisable in the sense that it degrades the true reliability of the scale. Fourth, none of these averages is a safe guide as to what any one real study will observe. We show that there is a divergence between α and true reliability of a scale when items are dropped from a scale, and that researchers who draw different samples from the same population cannot know whether they are removing an item based on signal of its poor reliability, or noise.

4.1. What we expected, and what we found

Because this simulation was not preregistered, the expectations recorded in the Method represent our broad intuitions prior to running the simulation rather than precisely preregistered predictions.

Three expectations were confirmed. The full-pool calibration bias was small and negative, and shrank toward zero with N at every item pool size. The structural gap was negative in every one of the 25 conditions. Prediction intervals were wide relative to the means in conditions with short scales and small samples, narrowing as either design factor increased.

One expectation was clearly wrong, and its disconfirmation is informative. We had expected selection optimism to increase with item pool size, on the reasoning that more candidate items in a given scale yields a larger expected maximum over noise. In fact, the opposite is the case. The bias against the population α of the retained items is largest at $k = 5$ (0.0435 at $N = 20$) and is already slightly negative at $k \geq 15$ with small samples. Our original reasoning counted the number of candidate items in a scale but not their individual leverage. As the pool of items grows, each contributes proportionally less to α, so the maximum is taken over more items with smaller individual effects, and the second effect dominates. At longer pools it is further offset by the small-sample downward

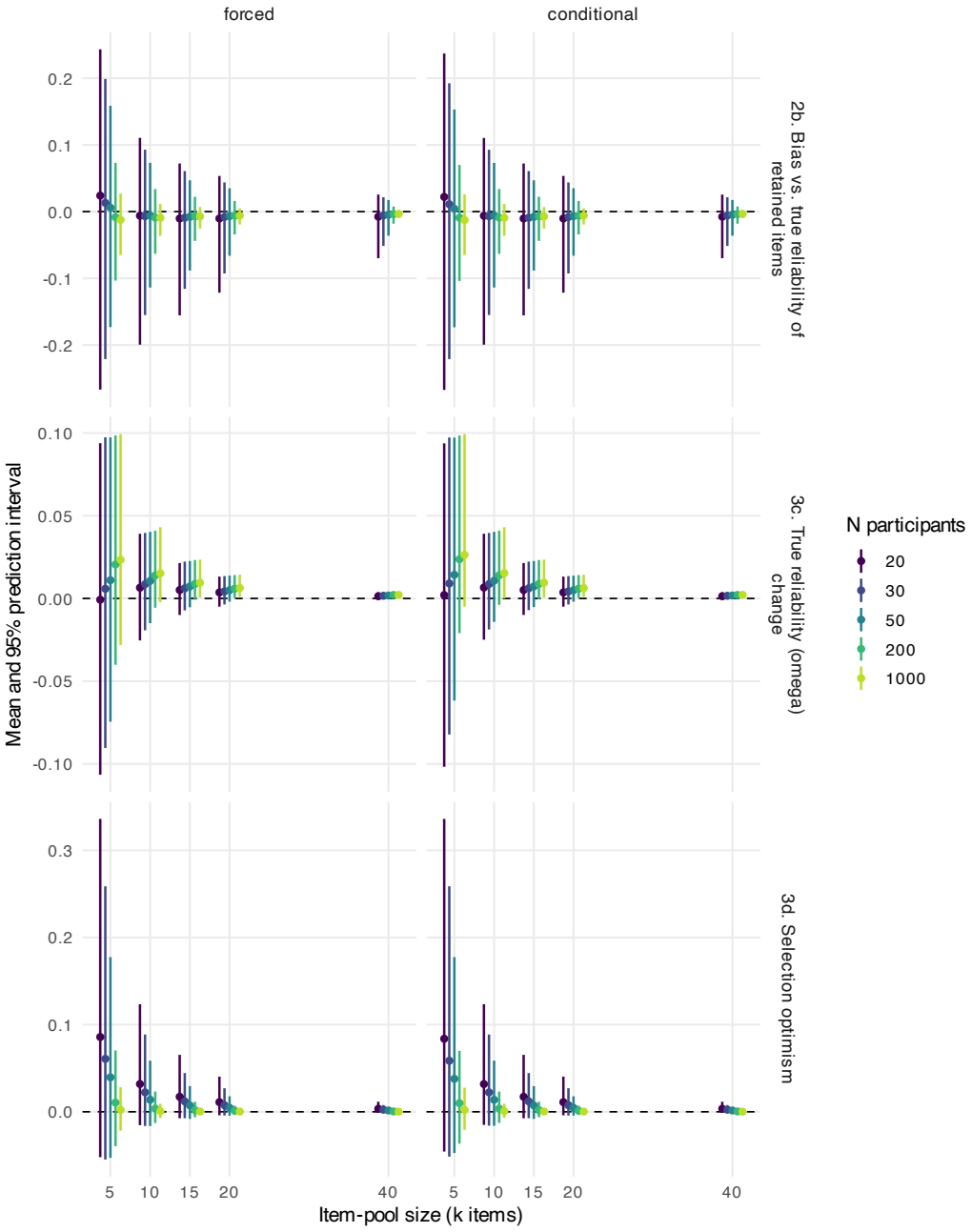


Figure 5: Mean and 95% prediction interval for the bias in the reported alpha against true reliability (top), the true reliability change (middle), and the selection optimism (bottom), for the forced (left) and conditional (right) strategies, by item-pool size and sample size. Points are means, bars span the empirical 2.5th to 97.5th percentiles across replications. Bars are dodged horizontally within each item-pool size so the five sample sizes are distinguishable. Note the differing y-axis scales between rows.

calibration bias seen in Cronbach's α, working in the opposite direction. The practical implication runs opposite to the original intuition, in that longer scales are not more exploitable by this rule, but less so.

A second expectation was right in direction, but overstated. We had expected the reliability-scale bias to be negative at large N and possibly positive at small N, and treated the location of the sign change as the open question. In fact, it is positive in only 3 of 25 conditions, all at the shortest pool size. The sign change is present, but it is confined to a narrow corner of the design. Building the results around this reversal, as an "alpha inflation" account would, would have overstated the scope of this finding. Across most of the grid the reported number is only slightly wrong, and had we stopped at the estimands we thought would carry the paper, we would have concluded that item dropping is a minor problem. The harm rate, for which we deliberately made no directional prediction because we could not anticipate the balance between genuinely bad items and pure length penalties, is what shows otherwise. Up to 49.6% of replications made worse. The result we did not predict is the one that matters for practical impact, and it was captured only because we included an estimand we included precisely because we did not know what it would show.

Finally, Pr(drop) rose with item pool size as expected, but its behaviour in N was not anticipated and proved to be the more informative pattern. At short pool sizes it falls as N increases, from 91.0% to 77.9%, because noise-driven drops become rarer with sampling precision and only genuinely harmful items are left to remove. The rule therefore fires most often exactly where its firing means least.

4.2. Reconciling with the simulation literature

In previous work, smaller simulation studies have been conducted that investigate the practical impacts of item dropping. Kopalle & Lehmann (1997) examined the effect of selecting scale items based on observed correlations in the same sample used to report the α of the scale, framing the resulting overestimation in α as "alpha inflation". They argue against data-driven item selection without theoretical justification or cross-validation, showing that the biased α values are most frequently seen when sample sizes are small, when a large number of items are dropped, and when inter-item correlations are low. Their simulation was constrained to two sample sizes, three item pool sizes, and 50 replications per cell, representing a considerably smaller simulation than the present study. Kopalle & Lehmann (1997) dropped items based on factor loadings from an EFA and define bias relative to the α that would result from retaining the same number of items chosen at random. This benchmark captures over-conditioning on the sample at hand, but does not speak to how well the reported α relates to the true reliability of the measure. By contrast, the present simulation drops items based on α-if-item-removed, and evaluates bias against two distinct targets: the population α of whichever items were retained, and the true reliability of those items. The present study demonstrates these findings robustly, extending beyond those of Kopalle & Lehmann (1997).

The divergence between α and true reliability, even with no item selection at all, is consistent with a long-running critique of Cronbach's α. Sijtsma (2009) argued that α cannot in general equal the true reliability of a scale given its inter-item covariance structure, a point developed in the α/ω literature (McDonald, 1999; Revelle & Zinbarg,

2009; Zinbarg et al., 2005), which shows the two coefficients are equal only under the restrictive condition of essentially equal item loadings. Our structural-gap result is a direct simulation demonstration of that theoretical point under an empirically calibrated loading distribution, and shows its magnitude is not negligible relative to the biases usually discussed in the item dropping literature. At the shortest scale examined it is 0.024, which exceeds the selection optimism observed anywhere in the design outside the $k = 5$ column.

4.3. Bias in the reported number versus harm to the scale

The most consequential finding is not that the reported reliability number is wrong. Across most of the design it is only slightly wrong, with magnitudes mostly under 0.01. It is the decision itself to drop items that is frequently detrimental. In the worst condition, the forced item drop rule reduced true reliability in 49.6% of replications while its mean apparent in-sample gain was 0.090. The conditional item drop rule, which acts only on an apparent improvement, still did so in 46.7% of the replications in which it acted.

A modest average bias can coexist with a near-chance probability of an outcome that reduces the reliability, because the bias measure averages over both helpful and harmful instances of item dropping while the harm rate isolates how often the harmful case occurs. Nothing about the reported α itself distinguishes the two scenarios. This bears on how the downstream result should be read. Stefan & Schönbrodt (2023) show that item dropping on α inflates the false positive rate of the subsequent hypothesis test to as high as 30%, and it would be natural to attribute that to the reported reliability of a measure being overstated. The present results suggest that is largely not the mechanism: instead, across most of the design, the reported α is close to correct. In the majority of conditions, it is too low rather than too high.

What item dropping entails is a decision point, after seeing the data, at which the researcher may retain either of two scales that are almost indistinguishable on the available evidence. The inflation of α via item dropping is therefore better understood as a researcher degree of freedom than as a mis-estimated reliability. This distinction is valuable, because the two call for different remedies: reporting ω instead of α would address the second, and not the first. This is also why the average is the wrong summary to lead with for practice, as a mean Δ_ρ of -0.0007 at $k = 5$, $N = 20$ implies there is "no effect", but in reality is the average of a distribution in which roughly half of studies are made less reliable and roughly half are made more reliable.

4.4. Mean bias understates practical risk

A mean bias near zero is compatible either with every study landing close to zero, or with studies scattered across a wide range that happens to average around zero. Table 7 shows the design is unambiguously the second case for short scales: the $k = 5$, $N = 20$ cell alone spans [-0.267, 0.243]. Additional Monte Carlo replications would narrow the standard error on the mean further, but would not narrow this interval by anything, since it is a property of the distribution rather than of how many times we sampled from it. Reporting means with Monte Carlo standard errors alone, which is common practice in simulation studies, would understate what item dropping can do in a single study by a factor of roughly 392 here, and by a larger factor still in any simulation run with more replications.

4.5. Practical implications

In short scales and small samples, precisely where the reported number is most likely to be optimistically biased, the decision to drop a scale item is most likely to be wrong. Prior simulation work have shown that unbiased estimation of α at small N is possible when the largest eigenvalue of the item covariance matrix is large (Yurdugül, 2008), indicating that bias in α depends on how strongly unidimensional the items are as well as on sample size. Every pool generated in the current simulation is unidimensional by construction (see Limitations), so this moderator could not vary within our design; scales with weaker general factors than ours would likely fare worse, not better.

The key distinction for measurement practice is between what is knowable before a study is run and what is knowable after. Scale length and sample size are properties of the study design. Everything this simulation quantifies as risky, that is, the optimistic bias, the harm rate, and the width of the prediction interval, is a function of those two design choices and of nothing about the data subsequently collected, holding the loading distribution fixed at the trait-scale-typical one simulated here. The risk can therefore be known in advance for scales resembling the corpus used here. A researcher could take a proposed k and N, find the corresponding row of Table 5, and know before collecting data roughly how much to trust a future item dropping decision from that study. This is an unusual property for a bias to have; many statistical biases depend on unknown population parameters that the researcher cannot inspect in advance. This one depends only on design choices that are already written down in the preregistration or method section, together with the assumption that the scale at hand is exchangeable with the trait-scale pool that calibrated the simulation. A scale with unusually strong and uniform loadings faces lower risk at the same k and N, and a weaker one higher (see the clustering limitation below).

Psychological measures are overwhelmingly used once and then abandoned: 43% appear only a single time in the APA PsycInfo record of tests used in studies (Elson et al., 2023), and 79% are not reused more than twice (Anvari, Alsalti, Oehler, Hussey, et al., 2025; Anvari, Alsalti, Oehler, Marion, et al., 2025). For most measures there is therefore never a second sample in which an apparent in-sample gain could fail to replicate, and a researcher may be overfitting considerably more than they realise. Understanding the risk profile of a scale in advance of data collection, and classifying scale adjustment as a researcher degree of freedom, could reduce the likelihood of this single-use model of adjusted measures and improve interoperability of measurement scales across psychology.

The post hoc decision-rule analysis sharpens this advice. Where a large sample is available, a simple threshold works: act on an alpha-if-item-removed recommendation only when the apparent gain exceeds .02, which at $N \geq 200$ kept the probability of harming the scale at or below 7.0% wherever the rule recommended dropping at all. Where the sample is small, no threshold examined made the decision safe, so the recommendation is to not act at all and to treat the table as uninformative.

This gives a partial answer to the question Hussey et al. (2025) raise, of how a researcher can tell a genuinely poor item from an artefact of their own sample without a second sample to check against. The answer these results support is that they cannot tell from the analysis, but they can bound how often they would be wrong from the design. At

$k = 40$ with $N = 1000$ that bound is negligible. At $k = 5$ with $N = 20$ it is close to a coin flip.

What cannot be known in advance is anything from inside the alpha-if-item-removed table itself. The table that `psych::alpha()`, or any equivalent, like SPSS, produces looks exactly the same whether the item it nominates is genuinely the weakest link in the scale or is the item that happened to draw an unfavourable sample of participants this time. Two researchers who drew two different samples of the same size from the same true scale would each see a complete, definitive-looking table, possibly nominating different items for removal, and each would have exactly the same apparent justification for acting on it, even though at most one of them has observed real signal. The safeguard has to come from knowing where k and N sit in a risk table such as Table 5, because nothing internal to the analysis will indicate the risk level.

4.6. Limitations

The main limitation is that both strategies consider dropping at most one item, in one step. Neither iterates nor targets a final scale length, so multi-item dropping is not simulated but is instead noted as a future direction below.

This simulation shows multiple consequences of item dropping when reliability of a scale is under consideration, but says nothing about whether an item dropping recommendation would *replicate* in practice. Each replication draws one sample per pool, so the design can estimate how often a drop harms true reliability, but not whether the same item would be nominated again in a new sample from the same scale, nor whether an apparent in-sample gain would survive out of sample. A recommendation replicates exactly when it is driven by the stable, scale-level component of the loadings rather than by sampling noise, and that is precisely the component the marginal fit flattens away. A future simulation that parametrises the loading distribution hierarchically, so that scales genuinely differ from one another, and that splits each pool's data into training and testing samples, would investigate this point. Nominating the item in the training sample and assessing the decision's out-of-sample performance in the testing sample would examine whether the same item is nominated again, whether the reported α holds up, and whether the apparent gain in reliability is realised. The Future directions below sketch this design.

The population targets in the present study are realised rather than idealised subset targets. That is, they are the true properties of whichever items a given replication's procedure retained, not of the best possible subset. This is the right target for the applied question, but it means the targets vary by strategy and by replication rather than being a single fixed value per condition, a deliberate departure from the classical estimation setup that we flag so it is not mistaken for an oversight. Relatedly, only sample α is evaluated as the reported estimator. Sample ω is never estimated from data anywhere in this design, it is used only as a population-level target.

The 95% prediction intervals conflate two sources of variation, a fresh item pool and a fresh sample in each replication, so they support a field-level claim about studies of trait scales in general rather than a claim about resampling from one fixed scale. Separating the two would require a nested design with several samples per drawn loading vector. This is the same structure that the item-choice replicability question below requires. As such, the two are deferred to a single follow-up study.

Finally, data-generating processes vary considerably between simulation studies, and identifying where ours limits our conclusions is worthwhile despite its being empirically grounded rather than speculative. The corpus used to fit the loading distribution consists of individual-differences and personality measures, which may differ considerably from the scales on which item dropping decisions are made in clinical psychology or in contexts where psychometric properties carry more consequential weight.

Two idealisations of the generated data itself also bound the generality of the results. First, items are continuous and multivariate normal, whereas almost every α in the motivating literature is computed on ordinal Likert responses. Categorical items attenuate inter-item covariances, α computed on them behaves differently at small samples, and item dropping could interact with asymmetric response thresholds in ways a continuous data-generating process cannot exhibit. Second, every generated pool is exactly unidimensional: the single-factor model that α presupposes, short of tau-equivalence, is true by construction, so the only kind of genuinely weak item that can exist here is one with a low or negative loading. In practice, alpha-if-item-removed nominations are often driven by multidimensionality, cross-loadings, or correlated residuals, cases in which the nominated item is a signal of model misspecification rather than of noise, and in which removing it changes what the scale measures rather than merely how precisely. Our conclusion that the table cannot distinguish a weak item from a sampling artefact therefore describes the *best case* for the practice, the one in which the model the table implicitly assumes is correct. Simulation designs that build in realistic indicator distributions and deliberate misspecifications, in the manner of Greene et al. (2019), who simulated binary and skewed indicators under a range of structural misspecifications precisely because fit-based model comparisons behave differently there, would test whether the harm rates reported here are under- or over-estimates for real data.

More consequentially, the empirical loading distribution is a marginal, independent-item fit: each item's loading is drawn i.i.d. from the one pooled distribution, ignoring which items belong to which scale. Real loadings cluster within scales. Some scales are uniformly well constructed, while weak and negative loadings concentrate within particular, often multidimensional, scales. Under the i.i.d. fit that clustering is erased as genuinely weak items are sprinkled uniformly at random across generated pools, every sufficiently long pool contains some droppable item, and no pool is systematically better constructed than any other. The fit reproduces the marginal distribution of loadings faithfully, but it understates how much pools differ from one another, because clustering concentrates item quality within scales and the i.i.d. draw spreads it evenly. This matters for the estimands most central to this article, which is that the harm rate and the probability of dropping both turn on whether the nominated item is a stable property of the scale or an accident of the sample, and a data-generating procedure that understates between-scale heterogeneity misstates the balance between those two components, and with it how the marginal harm rate is composed.

4.7. Future directions

The replicability question raised in the limitations could be investigated with a same-item agreement measure benchmarked against a uniform chance floor rather than against Cohen's kappa, and with a paired training/testing design that distinguishes whether the same item gets nominated twice from whether a fixed item's apparent gain survives out of

sample. Because that design draws several samples from each item pool, it would also decompose the harm rate reported here. A marginal harm rate of close to one half is compatible with two quite different worlds: one in which every short scale is a coin flip, and one which mixes scales with no genuinely weak item, where dropping is near-certain to harm, with scales containing one, where it is almost certain to help. These carry opposite practical advice – the present design cannot distinguish them, but the nested follow-up can. Combining that analysis with the bias and harm-rate results here would complete the picture this pair of simulations is building toward.

A second extension would make the generated data more realistic on the two dimensions identified in the Limitations: ordinal indicators, generated by thresholding the latent responses at empirically calibrated cut points, and deliberately misspecified structures, such as minor secondary factors, cross-loadings, or correlated residuals, following the approach of Greene et al. (2019) of simulating realistic indicator distributions under a range of structural misspecifications. The misspecified case is the more informative one, because it is the only setting in which an alpha-if-item-removed nomination can be *right for the wrong reason*, flagging an item that genuinely does not belong while the analyst's unidimensional reading of the scale is itself in error.

A third extension would evaluate sample ω as an alternative reporting statistic, since the present study uses ω only as the population target against which sample α is judged rather than as a competing sample-based estimator. A fourth would add an external criterion, which would let the validity consequences of dropping be estimated directly rather than inferred from the reliability change. Finally, whether the specific practices simulated here reflect how item dropping actually occurs in the published literature remains an open empirical question.

4.8. Conclusion

Item dropping to improve Cronbach's α introduces bias into psychological measurement in ways that depend on the scale length and sample size. In short scales and small samples, item dropping reduces true reliability in close to half of simulated studies, including when the rule acts only on an apparent in-sample improvement. In longer scales and larger samples, it produces the opposite distortion, making the scale's reliability appear slightly lower than it truly is. Neither distortion is visible from inside the analysis that produces it. What is visible in advance is the study design, and researchers planning short scales with small samples should treat an alpha-if-item-removed recommendation from such a study as close to uninformative. In longer scales with more items, the true reliability change is close to zero, making item dropping unlikely to help reliability. Item dropping can therefore be understood as a researcher degree of freedom that ranges from actively lowering the reliability of a scale, to one that makes an essentially inconsequential change to reliability while over-conditioning on the sample at hand.

Author notes

CRediT authorship contribution statement

ZF: Writing – original draft. IH: Conceptualization, Methodology, Software, Formal analysis, Data curation, Visualization, Writing – review & editing.

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Conflict of interest

The authors declare that they have no conflicts of interest.

Data and code availability

All simulation code and results, including the full ADEMP specification, the validation of the reliability functions, per-condition tables for every estimand, and Monte Carlo standard errors for every estimate, are available at https://github.com/ianhussey/replicability-of-item-dropping.

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